Merits:

- The value of standard deviation is based on every observation in a set of data.
- It is less affected by fluctuations of sampling.
- It is the only measure of variation capable of algebraic treatment.

Limitations:

- Compared to other measures of dispersion, calculations of standard deviation are difficult.
- While calculating standard deviation, more weight is given to extreme values and less to those near mean.
- It cannot be calculated in open intervals.
- If two or more data set were given in different units, variation among those data set cannot be compared.

Example 6.6

Raw Data:

Weights of children admitted in a hospital is given below calculate the standard deviation of weights of children.

13, 15, 12, 19, 10.5, 11.3, 13, 15, 12, 9

Solution:

$$\text{A.M.,} \quad \bar{x} = \frac{\sum x}{n}$$

$$= \frac{13 + 15 + \ldots\ldots + 49}{10}$$

$$= \frac{129.8}{10}$$

$$= 12.98$$

Deviation from actual mean

x	$d = x\text{-}12.98$	d^2
13	0.02	0.0004
15	2.02	4.0804
12	–0.98	0.9604
19	6.02	36.2404
10.5	2.48	6.1504

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